User Guidance for CHARLS-HCAP Wave 4

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This document provides guidance to users of the CHARLS-HCAP (Wave 4, 2018) data.

The CHARLS-HCAP battery used in CHARLS Wave 4 was translated into Mandarin from the English language version of the HCAP battery used in the US Health and Retirement Study. To maintain a high degree of harmonization with the Health and Retirement Study version of the HCAP battery, this translation was strict in nature without content adaptations for the cultural context of China, or for illiteracy and sensory impairment, which have a higher prevalence in China than in the United States.

All neuropsychological tests rely to varying degrees on written, visual, and auditory cues. Thus, illiteracy and sensory impairment can affect performance on neuropsychological tests that have not been appropriately adapted to account for these factors. The results of neuropsychological testing should be interpreted carefully for CHARLS-HCAP Wave 4 participants with these characteristics.

In addition, testing a person in a language or dialect that is not their primary language or dialect may put them at an unfair disadvantage. China is a linguistically diverse country in which Mandarin is widely and officially used across provinces and has been taught in schools since 1956. However, most provinces of China also use local spoken dialects of Mandarin and other languages, such as Cantonese and Hunanese. Many older adults in China do not fluently speak standard Mandarin or use it in their daily lives.

To reach participants, CHARLS-HCAP recruited interviewers according to their language or dialect proficiency and let them communicate with respondents in local dialects or languages. Word lists were translated into different languages such as Tibetan, Mongolian and Uygur in picture form beforehand, but other tests were communicated verbally by trained interviewers during the CHARLS-HCAP interviews. For spoken dialects of Mandarin that have no written form, interviewers administered the tests in local dialects. In situations where the interviewer did not speak the same language or dialect as the study participant, a local translator was informally recruited to translate during interview sessions at participants' homes. Local translators were not trained in neuropsychological testing or in the administration of the HCAP battery. This approach raises concerns about altered meanings of the HCAP test items and responses for CHARLS-HCAP participants who used translators or did not speak standard Mandarin, which could affect their measured performance on cognitive tests.

The following are recommendations intended to assist data users in making informed decisions with respect to analyses using the CHARLS-HCAP (Wave 4, 2018) data.

- **Understand the data**. We recommend data users read the CHARLS-HCAP documentation in detail to understand the content of the CHARLS-HCAP battery and how the CHARLS-HCAP data were collected. If possible, review the Mandarin version of the documentation which may contain additional details not available in the English translation of the documentation.

- **Examine the prevalence and patterns of non-numeric responses, such as "cannot answer", "don't know", refusal to respond, and non-assessment**. A variable may have a non-numeric value for a given individual in the dataset for a variety of reasons that may be difficult to discern, including cannot answer, don't know, refusal to respond, non-assessment, or other non-response, all of which comprise forms of missing data. We recommend that data users scrutinize the CHARLS-HCAP data for prevalence and patterns of these forms of missing data. For example, "cannot answer" could be due to the respondent not being able to answer a question due to cognitive impairment, not being able to understand the question due to vision or hearing impairment, not being able to respond due to illiteracy, or other reasons unrelated to cognition. Data users should examine for differential rates of and types of missingness across key analytical variables. What is done with this information will vary depending on the research question, key exposure and outcome being used in the analysis, and the research discipline.

- **Consider sensitivity analyses by literacy, language, and sensory impairment**. As a sensitivity analysis, consider sample restrictions to individuals who are literate, speak Mandarin, and/or do not have a visual or hearing impairment. Such restrictions ensure similar test conditions and levels of participant comfort with the testing conditions. However, depending on one's scientific question, restriction to these population subgroups, whether simultaneously or individually, may also obscure or induce biased associations relative to the general population of older adults in China.

- **Scrutinize individual cognitive tests**. As a sensitivity analysis or as the main analysis, include cognitive tests less sensitive to biases by confounders of an association under study. For example, if literacy might confound an association of interest, then the HCAP test involving writing a sentence, or composite scores incorporating this test, should be avoided. Or, if visual impairment may confound an association of interest, then tests involving visual object identification should likewise be avoided.

- **Consider implications for longitudinal analysis**. Many key questions of interest in aging are longitudinal in nature and involve the assessment of cognitive change. Several tests in the CHARLS-HCAP battery, including questions about orientation to time and place and a memory test, were also included in the core CHARLS survey in Waves 1, 2, and 3. Immediate and delayed word recall of 10 words is perhaps the most widely used test that is longitudinally available. It is important for users to be aware that the word list used in the CHARLS-HCAP in Wave 4 was different in terms of content and administration than the word list used in prior waves. Thus, observed change over time in word recall scores is spuriously steeper in slope at Wave 4 relative to change over time observed across prior waves. Simultaneously, the delayed word recall score is elevated in Wave 4 relative to previous waves because three, rather than one, immediate recall trials were administered. To address these form differences, we recommend use of equipercentile equated scores for memory scores in longitudinal analyses (the following paper explains the method: Gross AL, Inouye SK, Rebok GW, Brandt J, Crane PK, Parisi JM, Tommet D, Bandeen-Roche K, Carlson MC, Jones RN. Parallel but not equivalent: challenges and solutions for repeated assessment of cognition over time. J Clin Exp Neuropsy-

chol. 2012;34(7):758-72. doi: 10.1080/13803395.2012.681628. Epub 2012 Apr 30. PMID: 22540849; PMCID: PMC3574868.).